

ASSIGNMENT 4A

We begin by carrying out the integration in the Maxwell's lever rule

$$p(V_V - V_L) = \int_{V_L}^{V_V} p dV \quad (1)$$

where $p = \frac{RT}{V-b} - \frac{a\sigma}{V(V+b)}$

Carrying out the definite integration of the two terms using the formulas:

$$\int \frac{dx}{x-b} = \ln(x-b) + C \quad \int \frac{dx}{x(x+b)} = \frac{1}{b} \ln \frac{x}{x+b} + C$$

we obtain

$$\int_{V_L}^{V_V} p dV = RT \ln \frac{V_V - b}{V_L - b} - \frac{a\sigma}{b} \ln \frac{V_V(V_L + b)}{V_L(V_V + b)} \quad (2)$$

Combining equations (1) & (2) we obtain

$$p(V_V - V_L) = RT \ln \frac{V_V}{V_L} \cdot \frac{1 - \frac{b}{V_V}}{1 - \frac{b}{V_L}} - \frac{a\sigma}{b} \ln \frac{1 + \frac{b}{V_L}}{1 + \frac{b}{V_V}} \quad (3)$$

From equation (3) using $\frac{b}{V} = \frac{B}{Z}$ and $\frac{a\sigma}{bRT} = \frac{A}{B}$ we obtain

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SEQUENCE OF SECANT METHOD ITERATES

		$Z_v^0 = 1$			$Z_L^0 = 0$			
Tr	Pr	A	B	Zv	EOS	ZL	EOS	F(Pr)
0.7	0.1	0.104273	0.012377	0.899615	3.42E-08	0.017258	1.221E-08	0.119513
0.7	0.15	0.156409	0.018566	0.840523	2.23E-09	0.025848	-6.5E-07	0.464745
0.7	0.082691	0.086224	0.010235	0.918421	9.4E-08	0.014276	-5.96E-08	-0.05034
0.7	0.089269	0.093083	0.011049	0.91136	1.86E-07	0.015408	-1.06E-07	0.018559
0.7	0.087497	0.091235	0.01083	0.913272	1.55E-07	0.015103	-9.23E-08	0.000574
0.7	0.08744	0.091176	0.010823	0.913333	1.55E-07	0.015093	-9.19E-08	-6.7E-06
0.7	0.087441	0.091177	0.010823	0.913332	1.55E-07	0.015094	-9.19E-08	2.41E-09
0.7	0.087441							

$$\omega = -\log_{10} Pr - 1 \Rightarrow \omega = 0.05828$$

$$A = 0.42748 Pr/T_r^2 \cdot 1/\sqrt{T_r}$$

$$B = 0.08664 Pr/T_r$$

$$Z^3 - Z^2 + Z(A - B - B^2) - AB = 0$$

$$X_{k+1} = X_k - \frac{F(X_k)(X_k - X_{k-1})}{F(X_k) - F(X_{k-1})}$$

$$F(Pr) = (Z_v - Z_L) - \ln \frac{Z_v - B}{Z_L - B} - \frac{A}{B} \ln \frac{1 + \frac{B}{Z_v}}{1 + \frac{B}{Z_L}}$$

$$Z_V - Z_L = \ln \frac{Z_V}{Z_L} \frac{1 - \frac{B}{Z_V}}{1 - \frac{B}{Z_L}} - \frac{A}{B} \ln \frac{1 + \frac{B}{Z_L}}{1 + \frac{B}{Z_V}} \Rightarrow$$

$$Z_V - Z_L = \ln(Z_V - B) - \ln(Z_L - B) - \frac{A}{B} \left[\ln\left(1 + \frac{B}{Z_L}\right) - \ln\left(1 + \frac{B}{Z_V}\right) \right]$$

Rearranging we obtain

$$Z_V - \ln(Z_V - B) - \frac{A}{B} \ln\left(1 + \frac{B}{Z_V}\right) = Z_L - \ln(Z_L - B) - \frac{A}{B} \ln\left(1 + \frac{B}{Z_L}\right) \quad (4)$$

which is equivalent to saying $\phi_V = \phi_L$

ALGORITHMIC SOLUTION

Given: $T_r = 0.7$

Step 1. Guess two values of P_r (say $P_r^1 = 0.1$ & $P_r^2 = 0.15$)

Step 2. Calculate: $A = 0.42748 \frac{P_r}{T_r^2}$
 $B = 0.08664 \frac{P_r}{T_r}$

Step 3. Solve the cubic equation
 $Z^3 - Z^2 - Z(A - B - B^2) + AB = 0$ for Z_V and Z_L

Step 4. Substitute Z_V & Z_L in (4) to calculate the residuals say F_1 for P_r^1 and F_2 for P_r^2

Step 5. Calculate P_r^3 from the secant method
 $P_r^3 = P_r^2 - \frac{F_2 (P_r^2 - P_r^1)}{F_2 - F_1}$

Step 6. Repeat until convergence

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ASSIGNMENT 4B

Enthalpy & Entropy Calculations

We are given that $H(T=T_b, P=1atm)^{ref} = 0$

Therefore the enthalpy calculation @ any temperature T & pressure P is given by the following formula:

$$\text{Enthalpy @ } T \text{ \& } P = H(T, P) - H(T=T_b, P=1atm) + H^{ref} \Rightarrow$$

$$\text{Enthalpy @ } T \text{ \& } P = H(T, P) - H(T=T_b, P=1atm)$$

Reference States: Enthalpy=0 at the Normal Boiling Point

Temperature (C)	Pressure (bar)		Enthalpy (kJ/kmol)
	Saturated Liquid		0.00
80.04	1.01325	0.00321	57359.42

Temperature (deg C)	Temperature (deg C)	Pressure (bar)	Z	PR Density kmol/m3	Enthalpy kJ/mol	PR Enthalpy kJ/mol	Error
50	50	5	0.01666	11.1693	-4.2864	-3.9705	7.37%
60	60	5	0.01635	11.0423	-2.8846	-2.6776	7.18%
70	70	5	0.01606	10.9097	-1.4535	-1.3495	7.16%
80	80	5	0.01581	10.7710	0.007071	0.0141	99.88%
90	90	5	0.01558	10.6257	1.4969	1.4139	5.55%
100	100	5	0.01539	10.4733	3.0158	2.8505	5.48%
110	110	5	0.01522	10.3132	4.5636	4.3248	5.23%
120	120	5	0.01508	10.1444	6.1404	5.8381	4.92%
130	130	5	0.01497	9.9662	7.7466	7.3916	4.58%
140	140	5	0.01489	9.7774	9.3832	8.9872	4.22%
142.61	143.183	5	0.01487	9.7149	9.8148	9.5042	3.16%
142.61	143.183	5	0.90115	0.1603	36.703	36.5042	0.54%
150	150	5	0.90589	0.1569	37.634	37.3301	0.81%
160	160	5	0.91230	0.1522	38.906	38.5610	0.89%
170	170	5	0.91813	0.1478	40.196	39.8148	0.95%
180	180	5	0.92347	0.1437	41.504	41.0912	0.99%
190	190	5	0.92835	0.1399	42.83	42.3902	1.03%
200	200	5	0.93284	0.1362	44.175	43.7115	1.05%

ASSIGNMENT 4B

First we need to evaluate $C_p - C_v$. We know that

$$C_p - C_v = T \left(\frac{\partial P}{\partial T} \right)_V \left(\frac{\partial V}{\partial T} \right)_P \quad (1)$$

The Peng-Robinson equation of state is

$$P = \frac{RT}{V-b} - \frac{a\sigma}{V^2+2bV-b^2} \Rightarrow$$

$$\left(\frac{\partial P}{\partial T} \right)_V = \frac{R}{V-b} - \frac{d\sigma}{dT} \frac{a}{V^2+2bV-b^2}$$

Wolfram

$$\boxed{\frac{d\sigma}{dT} = -\frac{1}{T_c} \frac{k\sqrt{\sigma}}{\sqrt{T_r}}}$$

and therefore

$$\left(\frac{\partial P}{\partial T} \right)_V = \frac{R}{V} \frac{Z}{Z-B} + \frac{T_r}{T} \frac{k\sqrt{\sigma}}{\sqrt{T_r}} \cdot \frac{a}{V^2} \frac{Z^2}{(Z^2+2BZ-B^2)} \Rightarrow$$

$$\left(\frac{\partial P}{\partial T} \right)_V = \frac{R}{V} \left\{ \frac{Z}{Z-B} + \frac{k\sqrt{T_r}}{\sigma} \frac{AZ}{(Z^2+2BZ-B^2)} \right\} \quad (2)$$

We also calculate

$$\left(\frac{\partial P}{\partial V} \right)_T = -\frac{RT}{(V-b)^2} + \frac{a\sigma(2V+2b)}{(V^2+2bV-b^2)^2} \Rightarrow$$

$$\left(\frac{\partial P}{\partial V} \right)_T = \frac{RT}{V^2} \left\{ -\frac{Z^2}{(Z-B)^2} + \frac{2\sigma a}{RT} \frac{V^2(V+b)}{(V^2+2bV-b^2)^2} \right\} \Rightarrow$$

$$\left(\frac{\partial P}{\partial V} \right)_T = \frac{RT}{V^2} \left\{ -\frac{Z^2}{(Z-B)^2} + \frac{2A(Z+B)Z^2}{(Z^2+2BZ-B^2)^2} \right\} \quad (3)$$

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REVIEW OF DIFFERENTIATION FORMULAS

$$P = P(V, T) \Rightarrow dP = \left(\frac{\partial P}{\partial V} \right)_T dV + \left(\frac{\partial P}{\partial T} \right)_V dT \Rightarrow$$

$$0 = \left(\frac{\partial P}{\partial V} \right)_T \left(\frac{\partial V}{\partial T} \right)_P + \left(\frac{\partial P}{\partial T} \right)_V \Rightarrow$$

$$\left(\frac{\partial V}{\partial T} \right)_P = - \frac{\left(\frac{\partial P}{\partial T} \right)_V}{\left(\frac{\partial P}{\partial V} \right)_T}$$

$$\sigma = [1 + k(1 - \sqrt{T_r})]^2$$

$$\frac{d\sigma}{dT} = - \frac{1}{T_c} \frac{k\sqrt{\sigma}}{\sqrt{T_r}}$$

$$\frac{d^2\sigma}{dT^2} = \frac{1}{2T_c^2} \frac{k(k+1)}{T_r \sqrt{T_r}}$$

Combining equations (1) (2) & (3) and using

$$\left(\frac{\partial V}{\partial T}\right)_P = - \frac{(\partial P / \partial T)_V}{(\partial P / \partial V)_T} \quad [\text{see page}]$$

we obtain

$$\frac{C_p - C_v}{R} = \frac{\left\{ \frac{Z}{Z-B} + \frac{k\sqrt{T_r}}{\sqrt{\sigma}} \frac{AZ}{(Z^2 + 2BZ - B^2)} \right\}^2}{\frac{Z^2}{(Z-B)^2} - \frac{2A(Z+B)Z^2}{(Z^2 + 2BZ - B^2)^2}}$$

Now we must calculate the integral

$$\int_0^\infty \left(\frac{\partial^2 P}{\partial T^2}\right)_V dV = -a \frac{dT_r}{dT^2} \int_0^\infty \frac{dV}{V^2 + 2bV - b^2} \Rightarrow$$

$$- \int_V^\infty \frac{\partial^2 P}{\partial T^2} dV = +a \frac{k(k+1)}{2T_r\sqrt{T_r}} \frac{T_r^2}{T^2} \int_V^\infty \frac{dV}{V^2 + 2bV - b^2} \Rightarrow$$

$$- \frac{T}{R} \int_V^\infty \frac{\partial^2 P}{\partial T^2} dV = \frac{a k(k+1)}{RT} \frac{T_r^2}{b^2 T_r \sqrt{T_r}} \frac{1}{2\sqrt{2}} \ln \frac{Z+B(1+\sqrt{2})}{Z+B(1-\sqrt{2})}$$

$$- \frac{T}{R} \int_V^\infty \frac{\partial^2 P}{\partial T^2} dV = \frac{A}{B} \frac{k(k+1)}{\sigma \sqrt{T_r}} \frac{T_r}{4\sqrt{2}} \ln \frac{Z+B(1+\sqrt{2})}{Z+B(1-\sqrt{2})}$$

VBA CODE FOR THERMAL PROPERTIES USING PENG-ROBINSON EQUATION OF STATE

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Public Function Zprop(index As Integer, compCol As Range, T As Double, P As Double, Z As Double, optional P1 As Variant, optional P2 As Variant, optional BIP As Range = Nothing)
    'calculates the Compressibility Factor
    '-----
    'index Integer designating which value to return
    'compCol Range containing molar composition
    'T Temperature in C
    'P Pressure in bar
    'Z Compressibility factor
    'P1 Optional parameter for EOS
    'P2 Optional parameter for EOS
    'Zprop Range of properties: A,B,C, dz/dP, dz/dT, dz/dT
    '-----
    Call ABC(compCol, T, P, P1, P2, BIP)

    Dim Eq1 As Double, Eq2 As Double, Eq3 As Double, Eq4 As Double, Eq5 As Double, Eq6 As Double

    Eq1 = 1# / ((Z - B) + kappa * Sqr(Temp) / Sqr(sigma)) * A / (Z ^ 2 + 2 * B * Z - B ^ 2)
    Eq2 = -Z / ((Z - B) ^ 2 + 2 * A * (Z + B) * Z / (Z ^ 2 + 2 * B * Z - B ^ 2) ^ 2)
    Eq3 = -Eq1 / Eq2
    Eq4 = -Z * Eq1 ^ 2 / Eq2
    Eq5 = A / B * kappa * (kappa + 1) * Sqr(Temp) / sigma / 4# / Sqr(2#) * Log((Z + B * (1 + Sqr(2#))) / (Z + B * (1 - Sqr(2#))))
    Eq6 = -1 + Eq5 + Eq4

    Dim CV As Double, Cp As Double, Visound As Double, JT As Double
    Cp = Cpig(compCol, T) + Rg("SI") * Eq6
    CV = -Rg("SI") * Eq4 + Cp
    Visound = -Rg("SI") * (T + 273.15) * Cp / CV * Z * Eq2 * 1000# / MolWt(compCol)
    JT = (Eq3 - 1#) / Density(T, P, Z) / Cp * 100#

    If (Index = 1) Then Zprop = CV
    If (Index = 2) Then Zprop = Cp
    If (Index = 3) Then Zprop = Sqr(Visound)
    If (Index = 4) Then Zprop = JT

    End Function

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We are now ready to summarize the process of calculating the thermal properties of a fluid that follows the Peng-Robinson Equation of State

$$1. \left. \frac{T}{P} \frac{\partial P}{\partial T} \right|_V = \frac{1}{Z-B} + \frac{\kappa \sqrt{T_r}}{\sqrt{\sigma}} \frac{A}{Z^2 + 2BZ - B^2} \quad (\text{Eq. 1})$$

$$2. \left. \frac{V}{P} \frac{\partial P}{\partial V} \right|_T = -\frac{Z}{(Z-B)^2} + \frac{2A(Z+B)Z}{(Z^2 + 2BZ - B^2)^2} \quad (\text{Eq. 2})$$

$$3. \left. \frac{T}{V} \frac{\partial V}{\partial T} \right|_P = -\frac{(\text{Eq. 1})}{(\text{Eq. 2})} = \beta T \quad (\text{Eq. 3})$$

$$4. \frac{C_p - C_v}{R} = -Z \frac{(\text{Eq. 1})^2}{(\text{Eq. 2})} \quad (\text{Eq. 4})$$

$$5. -\frac{T}{R} \int_V^\infty \left[\frac{\partial^2 P}{\partial T^2} \right]_V dV = \frac{A}{B} \kappa(\kappa+1) \frac{\sqrt{T_r}}{\sigma} \frac{1}{4\sqrt{2}} \ln \frac{Z+B(1+\sqrt{2})}{Z+B(1-\sqrt{2})} \quad (\text{Eq. 5})$$

$$6. \frac{C_p^R}{R} = -1 + (\text{Eq. 5}) + (\text{Eq. 4}) \quad (\text{Eq. 6})$$

$$7. C_p = C_p^{IG} + R * (\text{Eq. 6}) \quad (\text{Eq. 7})$$

$$8. C_v = -R * (\text{Eq. 4}) + C_p$$

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RESULTS

Reference States: Enthalpy=0 at the Normal Boiling Point

Temperature (C)	Pressure (bar)	Enthalpy (kJ/kmol)
80.04	1.01325	0.00321
		57368.42

Temperature (deg C)	Temperature (deg C)	Pressure (bar)	Z	Enthalpy kJ/mol	PR Enthalpy kJ/mol	Gv kJ/(kmol K)	PR Cv kJ/(kmol K)	Error	Cp kJ/(kmol K)	PR Cp kJ/(kmol K)	Error	Phase
50	50	5	0.01666	-4.2864	-3.9705	97.389	102.14	7.37%	138.74	127.56	4.88%	Liquid
60	60	5	0.01635	-2.8846	-2.6776	100.06	104.66	7.18%	141.64	131.05	4.59%	Liquid
70	70	5	0.01606	-1.4535	-1.3495	102.86	107.13	7.16%	144.58	134.58	4.15%	Liquid
80	80	5	0.01581	0.007071	0.0141	105.68	108.67	99.88%	147.53	138.16	3.68%	Liquid
90	90	5	0.01558	1.4969	1.4139	108.44	111.97	5.55%	150.44	141.80	3.26%	Liquid
100	100	5	0.01539	3.0158	2.8505	111.12	114.33	5.48%	153.33	145.53	2.89%	Liquid
110	110	5	0.01522	4.5636	4.3248	113.71	116.68	5.23%	156.22	149.36	2.59%	Liquid
120	120	5	0.01508	6.1404	5.8381	116.21	118.94	4.92%	159.14	153.31	2.35%	Liquid
130	130	5	0.01497	7.7466	7.3916	118.63	121.18	4.58%	162.12	157.42	2.15%	Liquid
140	140	5	0.01489	9.3832	8.9872	120.99	123.37	4.22%	165.21	161.74	1.97%	Liquid
142.61	143.183	5	0.01487	9.8148	9.5042	121.60	124.06	3.18%	166.04	163.17	2.02%	Liquid
142.61	143.183	5	0.00115	38.703	36.5042	111.53	108.60	0.54%	125.35	120.37	2.72%	Vapor
150	150	5	0.00589	37.634	37.3301	113.19	110.26	0.81%	126.46	121.94	2.59%	Vapor
160	160	5	0.01230	38.906	38.5810	115.43	112.79	0.89%	128.10	124.23	2.29%	Vapor
170	170	5	0.01813	40.196	39.8148	117.68	115.29	0.95%	129.84	126.51	2.03%	Vapor
180	180	5	0.02347	41.604	41.0912	119.92	117.74	0.99%	131.67	128.78	1.82%	Vapor
190	190	5	0.02835	42.83	42.3902	122.16	120.16	1.03%	133.55	131.02	1.64%	Vapor
200	200	5	0.03284	44.175	43.7115	124.38	122.64	1.05%	135.47	133.24	1.48%	Vapor
								8.94%			2.73%	4.07%

$$9. v_{\pm}^2 = (RT) \cdot \frac{C_p}{C_v} \cdot Z \cdot (Eq. 2)$$

$$10. \mu = \frac{\beta T - 1}{\rho C_p} = \frac{(Eq. 3) - 1}{\rho C_p}$$

RESULTS

Temperature (deg C)	Temperature (deg C)	Pressure (bar)	Z	Sound Speed (m/s)	PR Sound Speed	Error	Joule- Thomson (K/bar)	PR Joule Thomson	Error	Phase
50	50	5	0.01666	1192.0	1007.68	15.46%	-0.038923	-0.0450	15.49%	Liquid
60	60	5	0.01635	1144.0	959.14	16.16%	-0.037061	-0.0421	13.48%	Liquid
70	70	5	0.01606	1096.4	912.54	16.77%	-0.035153	-0.0391	11.13%	Liquid
80	80	5	0.01581	1049.4	867.72	17.31%	-0.033185	-0.0359	8.29%	Liquid
90	90	5	0.01558	1003.4	824.51	17.83%	-0.031128	-0.0326	4.83%	Liquid
100	100	5	0.01539	958.31	782.76	18.32%	-0.028942	-0.0291	0.56%	Liquid
110	110	5	0.01522	914.07	742.33	18.79%	-0.026582	-0.0253	4.86%	Liquid
120	120	5	0.01505	870.53	703.06	19.24%	-0.023992	-0.0211	12.00%	Liquid
130	130	5	0.01497	827.54	664.83	19.66%	-0.021102	-0.0165	21.91%	Liquid
140	140	5	0.01489	784.86	627.47	20.05%	-0.017830	-0.0113	36.87%	Liquid
142.61	143.183	5	0.01487	773.77	615.74	20.42%	-0.016902	-0.0094	44.12%	Liquid
142.61	143.183	5	0.90115	198.87	198.94	0.03%	2.2072	1.7196	22.09%	Vapor
150	150	5	0.90589	201.79	201.40	0.19%	2.0381	1.6405	19.51%	Vapor
160	160	5	0.91230	205.56	204.92	0.31%	1.8393	1.5345	16.57%	Vapor
170	170	5	0.91813	209.13	208.32	0.39%	1.6687	1.4389	13.77%	Vapor
180	180	5	0.92347	212.55	211.61	0.44%	1.5211	1.3524	11.09%	Vapor
190	190	5	0.92835	215.84	214.81	0.48%	1.3926	1.2737	8.53%	Vapor
200	200	5	0.93284	219.00	217.93	0.49%	1.2800	1.2020	6.09%	Vapor
						11.24%				